



A Plagiarism Detector in Learning Management System for UTHM

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CHAPTER ONE

INTRODUCTION

1.1 Project Background

A Plagiarism Detector in Learning Management System (LMS) for UTHM, is to provide an integrated plagiarism detection tool for the learning management system that lecturers and students at Universiti Tun Hussein Onn Malaysia (UTHM) utilise. The goal is to help lecturers check student assignments' originality within the LMS system without relying on external technologies. The technology automatically compares the content of an assignment uploaded by a student to other submissions that have been stored, generating a comprehensive similarity report for lecturers. The initiative improves academic integrity and expedites the assignment review process by integrating this feature into the LMS.

According to Sozon et al. (2024), cheating and plagiarism have become serious problems in higher education institutions (HEIs). It affects educational quality as well as the reputation of higher education. In order to guarantee that students produce original work, plagiarism detection is a crucial part of contemporary academic institutions. Lecturers have historically relied on outside plagiarism detection software, which necessitates manual uploads, different logins, and occasionally extra fees. By incorporating plagiarism detection into the LMS, lecturers and students may have a more seamless experience while maintaining the security of all data in the university's digital environment.

Additional features like blockchain function, encrypted storage, secure authentication, and an analytical reporting module are included in the architecture of the suggested system. According to Wang et al. (2025), within the education sphere, blockchain applications protect academic records, ensure their trustworthiness, reduce fraud, and improve the value of educational assets. Every student's submission, plagiarism report, and grading record can be stored as a transaction on the blockchain. Once data is added, it cannot be altered or deleted without leaving a trace. This ensures that lecturers and students cannot modify submission dates or plagiarism scores dishonestly.

Additionally, the system's architecture promotes accountability and openness by enabling lecturers to quickly examine plagiarism reports and make knowledgeable assessments.

Ultimately, this project adds value through automation, data consistency, and user ease, which helps to improve the current LMS, AUTHOR. It offers a safe, effective, and user-friendly plagiarism detection tool while supporting UTHM's dedication to promoting academic achievement and innovation in digital learning environments.

1.2 Problem Statement

According to Turnbull et al. (2022), learning management systems form an integral part of the learning environments of most universities and support a wide range of diverse activities and operations. Nevertheless, despite their significance, there are still a number of issues that have an impact on academic institutions' efficiency, data integrity, and performance monitoring. These difficulties underscore the need for an enhanced system that guarantees confidentiality, transparency, and authenticity in academic contributions while also facilitating learning.

Universities like UTHM must implement cutting-edge resources that meet the demands of contemporary teaching and learning as higher education moves towards increasingly digital and automated processes. The institution's capacity to guarantee originality, data security, and efficiency is hampered by the existing LMS's lack of an integrated plagiarism detection capability. In addition to enhancing academic reputation, the implementation of an integrated plagiarism detector will show UTHM's dedication to developing educational technology. According to Hijriyanti et al. (2020), with the development of technology in education, of course, innovation in learning develops for the better.

The absence of crucial security features like activity tracking, and encrypted data storage is another major worry. Sensitive academic data, including as assignments, grades, and user credentials, is nonetheless susceptible to illegal access and data breaches in the absence of these. According to Almusfar et al. (2025), data security stands out as crucial because any weaknesses can result in breaches that erode user confidence. Through transparent, impenetrable record management, strengthening the LMS with improved security layers and blockchain hash integration can protect data and uphold academic integrity by detecting duplicates or tampered content.

Furthermore, the majority of LMS platforms currently in use have user interfaces that are frequently complicated or difficult to use, as well as restricted analytics capabilities. This makes it challenging for lecturers to adequately detect trends of academic dishonesty, identify learning gaps, and measure student progress. By using real-time analytics tools, lecturers can get useful information about student performance, interest levels, and plagiarism patterns, encouraging proactive academic support.

To sum up, these difficulties show how important it is to create a more sophisticated and safe LMS for UTHM. The goal of the suggested plagiarism detection system is to develop a dependable, open, and effective academic platform that upholds the highest standards of academic integrity while improving teaching and learning opportunities.

1.3 Objective

The objectives of this system are to:

1. Develop a Secure Learning Management System that includes an AutoPlagiarism Detection function to improve academic integrity.
2. Propose an enhanced platform with cutting-edge features designed specifically for UTHM lecturers and students, including performance metrics, real-time notifications, and transparent plagiarism reporting.
3. Test the system's efficacy, dependability, and usability through functional testing and user acceptance tests.

1.4 Scope

The design and development of a plagiarism detection tool within a Learning Management System (LMS) specifically for Universiti Tun Hussein Onn Malaysia (UTHM) is the main objective of this project. The system's goal is to offer a comprehensive, safe, and effective platform for handling academic assignments and guaranteeing student inputs are unique. Students can upload their assignments straight through the LMS, and lecturers can check the similarity report that is produced by the integrated plagiarism detection system. By removing the need for external plagiarism tools and guaranteeing that all operations are confined within a single, cohesive platform, this procedure helps optimise academic workflows.

The system scope includes data encryption to safeguard academic files, and user authentication for safe access in addition to plagiarism detection. Lecturers will have

access to alternatives for effectively managing assignments, comprehensive plagiarism reports, and data on student submissions. In the meantime, students can track deadlines with automated notifications, view submission statuses, and get feedback on their originality. According to Lo et al. (2012), adaptive web-based system can help the student obtain information in a form that fits the students' characteristics and fulfills the user's real needs automatically over the web. Web-based technologies including PHP, MySQL, HTML, CSS, and JavaScript will be used in the system's development to provide accessibility across a variety of devices and browsers. The project's overall scope includes developing, testing, and implementing a user-friendly, safe, and efficient LMS feature that enhances UTHM's digital learning environment and supports academic integrity.

Table 1: Users of System

User Type	Responsibilities
Lecturer	• Set the sensitivity of plagiarism detection for each assignment or course. • Get class-wide data and automated plagiarism reports. • Track metrics related to student behaviour, such as inactivity, late submissions, and recurring high similarity scores. • Upload teaching materials safely using watermarks or download limits. • Offer ideas for assignment or quiz grading. • Get notifications about possible problems with academic integrity, such as the detection of plagiarised content.
Student	• Get pop-up alerts about impending deadlines for assignments.

	• Send in assignments over an encrypted, safe upload channel. • Receive immediate comments or report on plagiarism and recommendations for appropriate citations. • Keep track of each submission's own plagiarism history. • Get learning insights and study reminders.
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Table 2: Functional Modules

MODULE	FUNCTION
User Authentication Module	Ensures safe access for administrators, lecturers, and students via role-based access control, two-factor authentication, and login credentials.
Course Management Module	Enables creation, editing, and organization of courses, assignments, lecture materials, and announcements.
Assignment Submission & Plagiarism Module	Records submission history, offers real-time originality reports, automatically checks for plagiarism, and enables students to turn in work safely.
Notification & Reminder Module	Sends emails and pop-up notifications in real time about impending deadlines for assignments, plagiarism alerts, and course announcements.
Performance Analytics Module	Creates class-level statistics, plagiarism trends, submission patterns, and student performance dashboards to help lecturers make data-driven decisions.
Communication & Collaboration Module	Enables safe assignment feedback, discussion boards, and messaging between students and lecturers.

Data Backup & Security Module	To avoid data loss or unwanted access, it makes sure that all system data, submissions, and reports are secured, safely stored, and routinely backed up.

1.5 Expected Outcome

A fully working Learning Management System with an integrated plagiarism detection feature that improves academic integrity and expedites assignment grading at UTHM is the anticipated result of this project. In order to facilitate equitable and effective grading, the system is supposed to automatically assess submitted projects for originality, produce precise similarity reports, and give lecturers comprehensive insights. Additionally, it will ensure that sensitive academic materials are shielded from unwanted access by strengthening data security through user authentication and encryption. The system attempts to lessen the effort of lecturers, improve feedback turnaround time and give students a more transparent and effective submission process by combining assignment management and plagiarism detection into a single platform. In the end, the initiative will help UTHM create a digital learning environment that is more safe, datadriven, and integrity-focused.

1.6 Project Significant

This project is important because it can improve academic integrity, security, and efficiency in the digital learning environment at UTHM. The initiative reduces expenses, protects data privacy, and streamlines the workflow for lecturers and students by integrating a plagiarism detection mechanism directly into the Learning Management System. This system encourages integrity and creativity in academic submissions while also streamlining assignment review. By adding secure authentication, automatic checks, and analytical tools to facilitate well-informed decision-making, it also fortifies the organization's digital infrastructure. All things considered, the project helps create a more

dependable, secure, and user-friendly academic platform that meets the changing demands of higher education.

1.7 Chapter Organization

This report contains six chapters as follows:

Chapter 1 describes the introduction of the project background, problem statement, objective, scope, expected result, methodology, project planning and project significant.

Chapter 2 reviewed all the relevant theoretical literature used in the proposed system.

Chapter 3 describes the relevant methodology for the proposed system.

Chapter 4 explains the analysis and design of the development system.

Chapter 5 explains the implementation and testing of the development system.

Chapter 6 describes the conclusion and future work for the proposed system.

1.8 References

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